

Introduction to Agile Software Engineering Minitrack

Ashley M. Aitken
School of Information Systems
Curtin University of Technology
ashley.aitken@cbs.curtin.edu.au

This minitrack seeks to explore the middle ground between traditional software engineering (TSE) and the new agile software development (ASD), or what we call agile software engineering (ASE). The minitrack includes six contemporary and high quality research papers.

The paper titled “Identifying Common Characteristics in Fundamental, Integrated, and Agile Software development Methodologies” by Dyck and Majchrzak presents research into the characterization of software development methodologies. It identifies common characteristics that can be found in both traditional and agile software development methodologies and describes a structured approach to the classification of these methodologies.

The paper titled “The Many Lives of an Agile Story: Design Processes, Design products, and Understandings in a Large-Scale Agile Development Project” by Read and Briggs presents results from a 2.5-year Action Research study of how an ASD team developing a complex software system adapted the user story concept and the Scrum agile management approach. It introduces and discusses the practicalities and advantages and disadvantages of a parameterized version of user stories.

The paper titled “Automated Acceptance Testing as Requirements Engineering Practice” by Haugset and Stalhane describes how the use of automated acceptance test-driven development (ATDD) impacts requirements engineering in software

development. In particular, it discusses how ATDD can be seen as a mix of the traditional documentation focus in requirements engineering and the agile focus on iterative communication.

The paper titled “Towards a Framework for Assessing Agility” by Poonacha and Bhattacharya addresses the need for organizations to be able to measure and monitor their agility. It identifies and categorizes the enablers of agility and then discusses how the analytic hierarchy process (AHP) and an adaptive network-based fuzzy inference system (ANFIS) can be used to calculate the agility of an organization (based on the values of various parameters entered for an organization).

The paper titled “Agile Values, Innovation and the Shortage of Women Software Developers” by Ken H. Judy discusses how the percentage of female US software developers has significantly declined over the years. It then considers if agile values and practices can drive changes in the software development industry to attract and retain more female developers.

The paper titled “Scrum Practice Mitigation of Global Software Development Coordination Challenges: A Distinctive Advantage?” by Bannerman, Hossain, and Jeffery examines whether Scrum practices, used in four global software development projects, provide any distinctive advantages in mitigating the coordination challenges found in such projects, when compared with traditional approaches.